

FORM 2

THE PATENTS ACT, 1970

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THE PATENTS RULES, 2003

COMPLETE SPECIFICATION

(See Section 10; rule 13)

TITLE OF THE INVENTION

AI-BASED DRUG SAFETY PREDICTION SYSTEM USING ANIMAL DNA AND

BIOLOGICAL DATA

APPLICANT

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The following specification particularly describes the invention and the manner in which it is to be performed.

AI-BASED DRUG SAFETY PREDICTION SYSTEM USING ANIMAL DNA AND BIOLOGICAL DATA

TECHNICAL FIELD

This invention falls under the fields of Artificial Intelligence (AI), Biotechnology, and Drug Testing. It is specially designed to help pharmaceutical researchers predict whether a newly developed drug is safe for a particular animal before physically testing it.

This system uses AI to make a smart guess about whether a drug is likely to be safe or harmful by looking at biological data from animals, like their DNA, blood values, and how their organs respond, and combining it with the drug's chemical structure.

BACKGROUND

Animal testing has been a common way to see how a drug works before giving it to people for a long time. This has helped find dangerous chemicals, but it also raises big ethical and scientific questions. Animals used in experiments often have unexpected reactions that can seriously hurt their health or even kill them.

This old-fashioned method is bad for animals and also costs a lot of money, takes a long time, and doesn't work well. Pharmaceutical companies put a lot of money into testing drugs that later fail because animal trials don't go well. The process also includes doing the same experiments on different types of animals, which adds to the time and cost of development.

Also, more and more people around the world want to cut back on or stop animal testing altogether. Different governments, research groups, and animal rights groups are telling scientists to find better and safer ways to do things, especially ones that use digital technologies like AI.

AI is currently being used in many drug development domains, primarily concentrating on human responses and forecasts. Few systems, nevertheless, are designed expressly to examine an animal's biological makeup and forecast a drug's compatibility using that information. The majority of tools ignore important animal-specific elements such as organ sensitivity, blood composition, and DNA sequencing.

By creating an AI-based system that accepts inputs from both drug molecular information and animal biological data, this invention seeks to close that gap. After that, it examines trends, contrasts them with historical findings, and generates a safety score, which aids researchers in determining whether a medication is suitable for testing on a specific species. This enhances the effectiveness and safety of early-stage drug trials while also saving animals.

OBJECTIVE OF THE INVENTION

The primary objective of this invention is to develop an AI-based system that can predict the safety of newly developed drugs for animals before any live testing is conducted. This system is designed to improve the efficiency, ethics, and accuracy of the drug development process by using biological and chemical data together.

KEY GOALS:

To create an AI engine that can process animal-specific biological data such as DNA sequences, blood test parameters, and organ sensitivity levels.

To analyze and compare that animal data with the chemical structure and ingredients of a new drug, identifying potential incompatibilities or toxic reactions.

To generate a clear output in the form of a suitability percentage score (e.g., 87.5%) and a status indicating whether the drug is “Safe” or “Unsafe” for the specific animal.

To reduce the number of live animal trials, avoiding unnecessary harm while saving time and cost during the early stages of drug testing.

To help researchers and pharmaceutical teams make informed decisions before proceeding to real-world animal testing or further development.

To provide a foundation that may later be expanded to support human safety predictions, based on genetic or biological similarities between species.

SUMMARY

This invention introduces an AI-powered system that helps predict whether a newly developed drug is safe for a specific animal species — before testing it on the animal. The system uses two major inputs: detailed biological data from animals, and the full chemical profile of the drug.

The biological data may include an animal’s DNA sequences, blood test values, and organ-specific sensitivity information. The drug-related data includes the chemical structure, ingredients, and any known toxicity markers. Once both datasets are entered, the system uses a trained AI model to analyze patterns and predict possible outcomes.

The AI processes these details and produces a suitability score (for example, “84.3% suitable for Rabbit”) along with a clear recommendation: Safe or Unsafe. This allows researchers to avoid unnecessary live testing, especially in early phases of drug development.

This solution not only makes drug testing more ethical and animal-friendly, but also helps pharmaceutical teams save time, reduce costs, and avoid failures caused by unpredictable animal reactions.

Additionally, the system is designed with flexibility in mind — so it can be extended in the future to support human prediction modules based on cross-species biological similarities.

BRIEF DESCRIPTION OF THE DRAWINGS

The following is the detailed description and accompanying drawings ,how we are going to test the process of drugs.

Fig 1 System Architecture Overview

This diagram presents the complete pipeline of the proposed AI system. It begins with two types of input: (1) animal biological data (DNA, blood biomarkers, and organ sensitivity levels), and (2) drug molecular data (e.g., chemical composition in SMILES format). These inputs are passed into a preprocessing module that standardizes and formats the data. The processed inputs are then fed into an AI model (e.g., Random Forest, Neural Network) which performs the core analysis. The final output is a percentage-based suitability score and a "Safe/Unsafe" label for the animal.

Fig 2 Animal Data Breakdown

This figure illustrates the different biological data types collected for each animal. Three main data sources are highlighted: • DNA: genetic sequence or gene markers of the animal. • Blood Biomarkers: enzyme levels (e.g., ALT, AST), hemoglobin, creatinine, and immune markers. • Organs: physiological data related

to liver, kidney, heart, etc. These components form the comprehensive profile of the animal for accurate drug compatibility analysis.

Fig 3 Drug Data Structure

This shows the structure of the drug-related input to the system. The drug profile includes:

- Molecular structure (usually in SMILES format)
- Active ingredients
- Known side effects
- Toxicity classifications from drug databases like DrugBank and Tox21.

These inputs are vectorized and used by the AI model to match with the animal's profile.

Fig 4 AI Model Flow

This diagram details the internal structure of the AI model used in the system. It includes:

- Input layer (for drug and animal data)
- Processing layers (dense layers for neural networks or decision splits in tree models)
- Output layer that gives:
 - A numerical suitability score (e.g., 84.7%)
 - A classification: Safe, Moderate, or Unsafe.

The model is trained using supervised learning on past animal-drug interaction datasets.

Fig 5 Output Display Format

It showcases a sample output interface/report. The result includes:

- Animal species (e.g., Mouse)
- Drug ID (e.g., D-021X)
- Suitability Score: e.g., 84.7%
- Final Status: SAFE for Animal.

This output helps researchers decide whether to proceed with physical animal trials or reformulate the drug.

Fig 6 Cross-Species Learning Model

This figure explains how the system is extended to learn across multiple animal species. For example, it can take data from mouse, rabbit, and monkey experiments to: Understand inter-species patterns, Improve accuracy of prediction

for a given species ,Identify drug behaviors that are consistently harmful or safe across species .This makes the system smarter and adaptable over time.

Fig 7 Optional Human Prediction Module

It presents an optional AI module that estimates human compatibility based on animal test data.It analyzes how closely an animal's biological response matches human genetic and biochemical data.The output is: Human Compatibility Score (e.g., 81.2% likely safe) , Risk markers detected or not.

DETAILED DESCRIPTION

This invention introduces a smart AI system that helps researchers predict whether a newly invented drug will be safe for a specific animal, even before giving it to the animal. This prediction is made by comparing two things:

1. The animal's biological data,
2. The chemical data of the drug.

Collecting Animal Data

Every animal has a unique body system — this includes its genes (DNA), blood test results, and how sensitive its organs (like liver or kidney) are to chemicals. The system collects this kind of information for each animal species. This data might come from lab test results, biological studies, or medical databases.

Understanding the Drug

Next, the system gathers full details about the drug. This includes: What the drug is made of (its chemical formula).The ingredients used in making it.Any known side effects or past test results (if available)

AI Makes the Comparison

Now, the smart part begins. The system takes the animal's biological data and the drug's chemical details and feeds them into an AI model (a machine learning program). This model has been trained using previous drug-animal case studies. So, it has learned how different types of animals react to different types of drugs.

The AI looks for hidden patterns, checks compatibility, and predicts how the animal's body might react to the new drug.

Getting the Result

After analyzing everything, the system gives a result in two forms:

A percentage score (e.g., "85.7% suitable for Rabbit")

A simple status: Safe or Unsafe.

This helps researchers know if it's okay to move ahead with real testing — or if the drug should be modified or rejected.

Why This Matters

This invention is useful because:

It can help avoid unnecessary harm or death to animals

It saves time and research cost

It gives early warning signs about drug safety

It supports ethical research

It increases the chances of success when drugs go for real testing later

In short, this invention helps researchers make smarter, safer, and more responsible decisions in the early stages of drug development.

Albert Francis

Application Agent (INPA - 4655)

CLAIMS

We claim that,

1. An AI-based drug safety prediction system for animals, comprising:

a data acquisition module, which collects animal-specific biological data including DNA sequences, blood biomarkers, and organ-related parameters for one or more species;

a preprocessing module, which normalizes and encodes both animal biological data and drug data to prepare for analysis;

an AI engine, trained to analyze and compare the processed animal data and drug data, and to generate a suitability score representing the predicted safety level of the drug for the target animal species;

a scoring module, which converts the AI output into a percentage score and classifies the drug as Safe / Unsafe / Moderate Risk based on threshold values;

a database, for storing animal biological data, drug profiles, AI model parameters, and prediction results for future reference and model improvement;

an optional extension module, configured to predict a probability-based human safety score by analyzing cross-species biological similarity data, without claiming clinical-level certainty.

2. The AI-based drug safety prediction system as claimed in claim 1, wherein the animal biological data includes but is not limited to genomic markers, blood test values (such as ALT, AST, creatinine), and organ tolerance data.
3. The AI-based drug safety prediction system as claimed in claim 1, wherein the drug data module utilizes cheminformatics tools like RDKit to convert chemical structure into numerical vectors suitable for AI analysis.
4. The AI-based drug safety prediction system as claimed in claim 1, wherein the AI engine comprises machine learning models selected from Random Forest, Neural Networks, Gradient Boosting, or equivalent classifiers trained on public datasets (such as Tox21, DrugBank, NCBI).
5. The AI-based drug safety prediction system as claimed in claim 1, wherein the scoring module assigns threshold-based classification to the drug, labeling it as Safe, Unsafe, or Moderate Risk for the target animal species.
6. The AI-based drug safety prediction system as claimed in claim 1, wherein the optional extension module estimates the probable human safety score based on cross-species biological similarity, serving as a preclinical advisory tool without replacing clinical testing.
7. The AI-based drug safety prediction system as claimed in claim 1, wherein the report generation module creates output reports in digital formats such as PDF, Excel, or web dashboards, showing species name, drug ID, suitability percentage, and recommendation.

8. The AI-based drug safety prediction system as claimed in claim 1, wherein the database stores encrypted animal biological data, processed drug data, and AI prediction results in compliance with data privacy regulations.

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ABSTRACT

AI-BASED DRUG SAFETY PREDICTION SYSTEM USING ANIMAL DNA AND BIOLOGICAL DATA

This invention presents an AI-powered system that predicts whether a newly developed drug is safe for a particular animal before performing any live testing. The system works by comparing detailed biological data of the animal—including DNA, blood test values, and organ sensitivity—with the chemical structure and ingredients of the drug. By using a trained machine learning model, the system analyzes patterns in how different animals react to different compounds.

It then generates a prediction in the form of a suitability percentage and a simple Safe or Unsafe label. This approach helps pharmaceutical researchers avoid harmful animal testing at early stages, reduce costs, and make faster, more ethical decisions during drug development.

Additionally, the system is designed in a way that it may optionally be expanded to estimate human safety, based on cross-species biological similarities. The invention promotes smarter and safer drug research, while supporting the ethical treatment of animals.